



Procedure to Measure End Play for the Fan Drive on 785C/D Engines

Maintenance and Repair

Application

Component Life Management

Component Renewal (CRC)

MARC Management

Procedure to Measure End Play for the Fan Drive on 785C/D Engines.....	1
1.0 Introduction.....	2
2.0 Best Practice Description.....	3
3.0 Implementation Steps.....	3
4.0 Benefits.....	6
5.0 Resources Required.....	6
6.0 Supporting Attachments / References.....	6
7.0 Related Best Practices.....	6
8.0 Acknowledgements.....	7

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1.0 Introduction

The cooling system performs several functions which are critical to proper machine operation: It maintains proper engine temperature for optimum performance, cools compressed inlet air to optimize combustion and dissipates excess heat from other machine systems and components.

Caterpillar has implemented hydraulically driven demand fan on certain products to improve fuel efficiency and provide more efficient cooling, but there are plenty of machines still in operation which use a mechanically-operated cooling fan.

A properly functioning fan drive is critical to engine reliability, as it works to maintain a safe temperature range. A failing fan drive bearings can cause complication in the repair due to fan and radiator core damage and result in expensive cost repairs.



Figure 1. Fan Drive system of 785C/D engine (source: media SERV1857).

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2.0 Best Practice Description

The fan hub assembly plays an integral roll in keeping the fan drive in place so it can maintain the optimal speed of the system. To prevent the fan drive from failing off, this Best Practice provides a procedure to measure end play for fan drive and accurately diagnose the problem and fix it as efficiently as possible.

3.0 Implementation Steps

1. Park the machine on a level surface and engage the parking brake;
2. In order to relieve the tension on the fan drive belts, Insert suitable 19.1 mm (0.75 inch) tooling 386-3359 into square hole in the belt tensioner, and pry the arm of the belt tensioner in a counterclockwise Direction;
3. Remove the belts;
4. Attach the magnetic base of a dial indicator to the support of the fan drive;



Figure 2. Position of magnetic base on fan drive Support.

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5. Place the indicator tip on the backface of the hub;



Figure 3. Place the indicator tip on the backface of fan drive hub.

6. Use tooling 4C-5647 to lever and move the hub;



Figure 4. Use a pry bar to move the hub in one direction until the movement stops.

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Procedure to Measure End Play for the Fan Drive on 785C/D Engines	DATE 1/27/2023	CHG NO 00	NUMBER 1.05-230127-1

7. Read the dial indicator movement;
8. If the dial indicator reading is above 0.25mm from point of reference at the zero position, replace the hub and bearings;
9. Install the belts;
10. Slowly release the arm of the belt tensioner in order to apply tension to the fan drive belts. Remove the tooling;
11. Take measurements every 500 hours PM and plot the measures in a graph to analyze the trend.

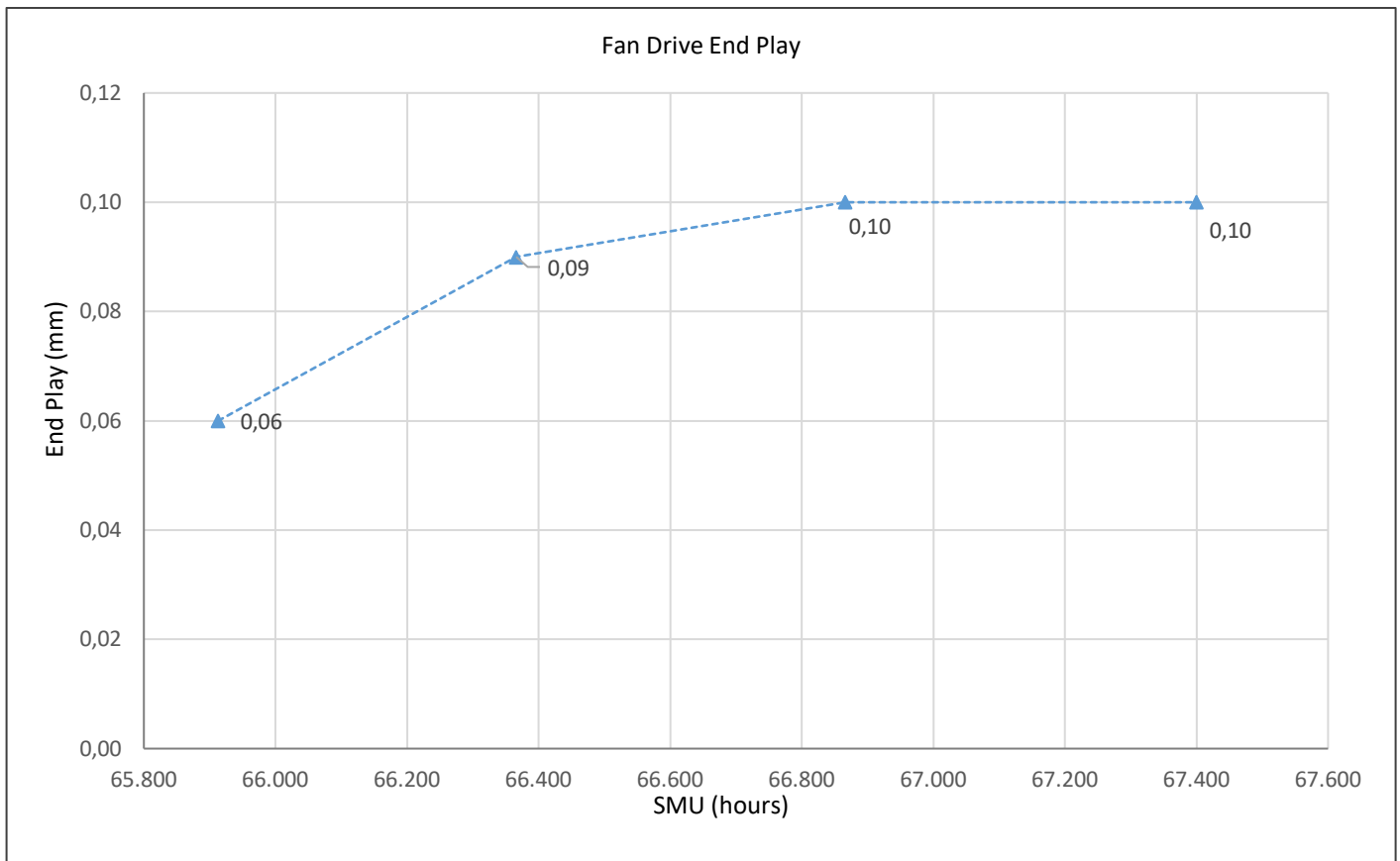


Figure 5. Graph for monitoring fan drive end play.

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4.0 Benefits

- Better parts inventory management;
- Monitoring fan drive end play;
- Avoid catastrophic failures and damages to the fan and radiator core;
- Reduction of unscheduled machine downtime.

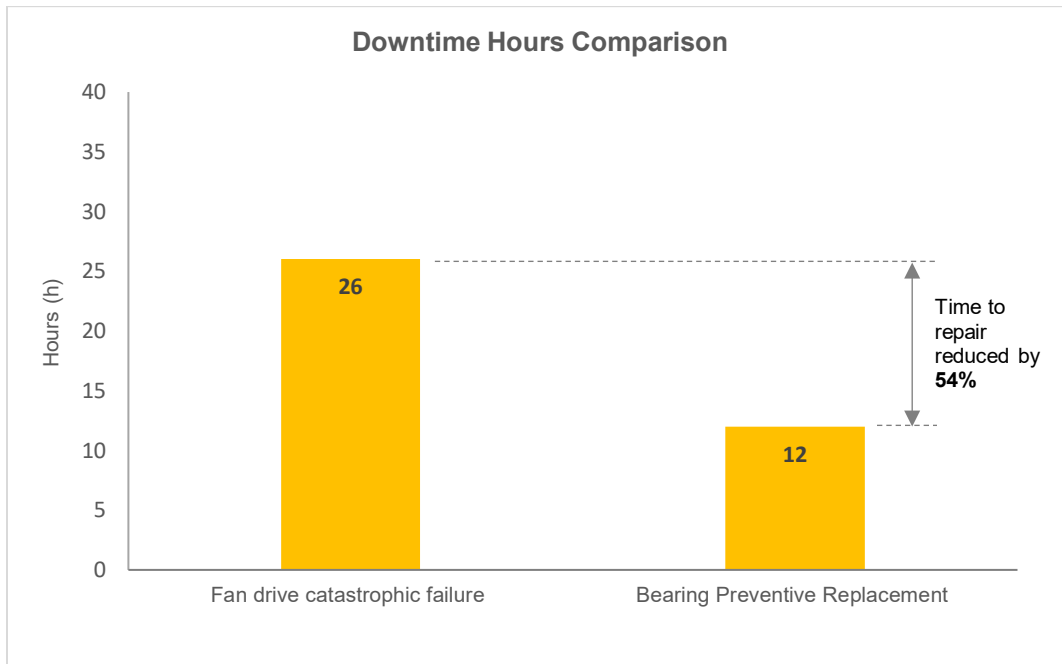


Figure 6. Downtime hours comparison.

5.0 Resources Required

- Dial Indicator
- Magnetic Base
- 4C-5647 Long Pry Bar
- 386-3359 Sliding T-Handle (3/4")
- Trained Technician

6.0 Supporting Attachments / References

Belts - Inspect/Adjust/Replace (SEBU7173-22)
 Fan Drive - Remove (SENR1486-03)

7.0 Related Best Practices

None

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